BrandGuard

Phase 1 | Evidence and product proposal

For the Inkloom AI brand-governance problem statement

Submitted by CODE CATALYST | Team leader: SANATH REDDY

Prepared 20 October 2026. BrandGuard is a working project name, not a claim of affiliation with Inkloom AI.

The proposal

BrandGuard is a proposed pre-publication review workspace that checks content against an organisation's approved guidelines, facts and assets. It explains each finding, links it to a source, suggests an editable correction, and routes the exact content revision to an authorised human reviewer.

Value proposition: Help creators resolve routine brand mistakes before review, while giving managers a traceable basis for approving content.

Problem and evidence

Growing teams distribute content creation across employees, agencies and AI tools. A guideline stored in a document does not ensure that a creator uses the current product name, an approved claim or the correct asset. Manual reviewers must repeatedly reconstruct those decisions.

External evidence supports investigating this problem:

- **Content consistency and approvals:** CMI's 2025 B2B report found that 42% of respondents cited creating content consistently as a challenge, and 33% cited workflow or approval-process challenges. The report covers 980 B2B respondents, mostly in North America, from a survey fielded in June-August 2024. These are historical, self-reported industry findings, not measurements of our proposed users. [S1]
- **Rules depend on context:** Mailchimp's published guidance distinguishes a relatively stable brand voice from tone that varies with the reader's situation. This supports evaluating tone with audience and channel context. [S2]
- **AI needs verification:** NIST's Generative AI Profile identifies confabulation and information-integrity risks. Our design inference is to ground findings in approved evidence and retain human judgement. [S3]

What Phase 1 establishes

This proposal defines the users, intended workflow, MVP boundary, architecture, responsible-design controls, research protocol and evaluation plan. The central hypothesis is that source-linked pre-checks can reduce repetitive review work without weakening human control.

Evidence status: Desk research completed. Interviews, usability sessions, product implementation and performance measurements have not been completed. All examples are fictional; all success thresholds are proposed targets.

01 | Users and validation

Initial organisational setting

The proposed first pilot is an English-language marketing team at a growing B2B organisation, working with an external creator or agency. This is a recruitment hypothesis, not an existing customer. Start with one brand, a small approved knowledge base and social posts, email copy and simple campaign graphics.

User	Current need	Proposed support
Creator or agency contributor	Understand what to change without losing the intended message	Inline findings, evidence links, editable suggestions and revision comparison
Brand manager	Apply current guidelines consistently across submissions	Versioned rules, review queue and recurring-issue summaries
Authorised reviewer	Check claims and record a defensible decision	Source-backed findings, comments, approval and reasoned exceptions
Workspace administrator	Control who can see material and approve it	Membership, permissions, source activation and processing settings

Research to conduct

Recruit **six consenting participants**: three creators, two brand managers and one authorised reviewer, ideally across two teams. Conduct a 20-minute interview and a walkthrough of one recent, sanitised content review per participant. Do not request confidential client materials without permission.

Ask these questions before showing the proposed solution:

1. Walk through the last item you submitted or approved. Who reviewed it?
2. Where did you find the applicable guidelines and approved facts?
3. What caused revisions, and which feedback repeated earlier feedback?
4. How did you resolve outdated or conflicting information?
5. What information could an automated service process, and who decides?
6. What evidence would make you accept or reject an automated finding?

Record participant code, role, consent, workflow steps, observed friction, review duration where measurable, and anonymised supporting notes. Separate direct observation from participant opinion and researcher interpretation.

Decision gates

Proceed with the proposed scope if at least four participants independently describe recurring guideline or evidence-related rework, and two teams can identify a person responsible for keeping sources current. These are recruitment-stage decision rules, not statistically representative market validation.

If source ownership is absent, prioritise guideline curation before automated checking. If review noise is the main concern, prioritise precision and finding dismissal over additional detection categories.

Next evidence artifact: An anonymised research log with actual observations, contradictions and resulting design changes. No participant quotes or interview outcomes are invented in this proposal.

02 | Product scope and feasibility

Planned MVP

Capability	Initial implementation boundary
Approved brand knowledge	Manually define or import text-based PDF/DOCX guidelines; a manager verifies extracted rules before activation. Store terminology, dated facts and original PNG/JPEG assets.
Content intake	Pasted text, text-based PDF/DOCX, and PNG/JPEG campaign graphics with creator-supplied context and alt text. Reject unsupported or unreadable inputs explicitly.
Text checks	Exact terminology rules and structured fact checks; evidence-grounded AI suggestions for tone, messaging and potentially unsupported claims.
Visual and accessibility checks	Approved-asset identity checks, image dimensions, declared colour-pair contrast, missing alt-text fields and OCR-assisted text review.
Explainable corrections	Finding location, category, severity, source version, rationale, proposed edit and evidence status. Changes require creator action.
Human workflow	Comments, revision history, assigned review, changes requested, approval and logged exceptions.
Governance insights	Confirmed findings by rule and category, dismissals, recurrence per reviewed item and sources with repeated ambiguity reports.

Boundaries that keep the product credible

OCR is an aid, not a guarantee of complete image reading. Exact asset matching can recognise an approved original file but cannot prove that a logo placed inside a flattened poster is undistorted. Poster logo placement, clear space and ambiguous colour extraction remain explicit manual-review tasks in the MVP.

Contrast checks use known foreground/background pairs. WCAG 2.2's minimum text-contrast guidance specifies 4.5:1 for ordinary text and 3:1 for qualifying large text, with exceptions. A raster image alone may not establish the displayed font size or background reliably. These checks do not establish overall accessibility conformance.

An unavailable or expired supporting source produces **unverified**, not “false.” Conflicting sources produce **needs source-owner review**. A model may suggest a tone improvement, but it must not present subjective preference as a proven factual violation.

Deferred beyond the MVP

Video/audio review, full design-file understanding, multilingual evaluation, automatic publication, broad internet fact-checking, enterprise SSO and production CMS integrations. Google Docs, Figma and CMS connectors are later integration candidates, not connected services.

Intended distinction

The design combines organisation-approved evidence, revision-specific review and feedback about unclear guidelines. Its value will be assessed against a manual guideline checklist; no claim of unique market coverage or superiority over existing products has been established.

03 | Workflow and demonstration

End-to-end journey

1. **Curate:** A manager adds rules, facts and assets, records their owner and validity dates, resolves conflicts, and activates a source version.

2. **Submit:** A creator selects brand, channel, audience and purpose, then supplies content. The interface displays the processing mode before analysis.

3. **Analyse:** The service checks supported rules, retrieves relevant approved evidence and produces located findings. Failed checks appear as incomplete.

4. **Resolve:** The creator inspects evidence, accepts or edits suggestions, or requests an exception. Every change creates a new revision.

5. **Review:** The assigned reviewer comments, requests changes, or approves the analysed revision after blockers are resolved or authorised exceptions recorded.

6. **Export and learn:** Export the approved revision and review record. Aggregate confirmed recurring issues to help the manager improve guidelines.

State path: Draft -> Analysing -> Needs changes / Ready for review -> In review -> Approved / Changes requested. Analysis errors return an incomplete state. An edit after approval creates a new draft requiring analysis and approval; it does not inherit approval.

Fictional example: LumaDesk

The following rules and content are invented for demonstration. They are not facts about an actual business.

Approved source	Synthetic rule or fact
G-01, terminology v1	Write the product name as “LumaDesk.”
F-01, trial terms v1	The trial lasts 14 days; no credit card is required.
G-02, claims v1	Performance or market-leadership claims require current approved supporting evidence.
G-03, tone v1	Use clear, calm language and avoid pressure-based calls to action.

Draft: “Lumadesk guarantees 10x productivity. Get your 30-day free trial. Sign up NOW!!!”

The expected review flags the spelling against G-01 and the trial duration against F-01. It marks “10x productivity” as unsupported under G-02 because no evidence is supplied. The call to action receives an advisory tone suggestion against G-03.

Suggested revision: “Try LumaDesk free for 14 days. No credit card required. Start your trial.”

The creator decides whether the suggestion preserves the campaign's intent. A reviewer still checks the final revision. A linked campaign graphic with no supplied alt text creates an additional accessibility-review task.

Review screen

Use a content preview with highlighted spans beside a finding panel showing evidence and proposed edits. Keep comments, version comparison and approval controls visible. Show outstanding blockers and unverified checks; avoid a single score that could imply guaranteed compliance.

04 | Architecture and technical decisions

Proposed data flow

Browser workspace -> authenticated API -> extraction and review worker -> deterministic checks plus approved-source retrieval -> optional model analysis -> validated findings -> human review -> approved export and audit record.

The API, worker, database and file storage remain within the chosen deployment boundary. An external model is a separate optional boundary, disabled until an administrator selects a provider and approves its data handling.

Component	Proposed choice and reason
Interface	React with TypeScript; content preview, accessible finding panels and explicit workflow states
Backend	Python FastAPI; shared parsing, rule execution and model-response validation
Storage	PostgreSQL for users, rules, revisions and decisions; private file storage for uploaded originals
Retrieval	Begin with structured filters and lexical search; add pgvector only if retrieval evaluation shows a benefit
Extraction	pypdf, python-docx and Pillow; Tesseract for optional OCR with visible extraction limitations
AI adapter	A locally hosted instruction model for the pilot if available hardware meets quality and latency gates; exact model and licence remain to be selected
Execution and tests	A persistent database job queue and worker; pytest for rules/access control, Playwright for critical browser flows; Docker Compose for reproducibility

These are planned choices, not installed product components. Pin versions and record licences at implementation. No external paid API or model has been selected or used to analyse brand content for this proposal.

Evidence-grounded review

Filter sources by organisation, brand, permissions, activation status and effective date before retrieval. Each review run snapshots the source versions used. Combine explicit checks with retrieved passages for semantic analysis; request structured findings referencing only supplied source IDs.

Validate cited IDs, quoted evidence and content locations against that snapshot. Structural validation cannot establish semantic correctness, so reviewers assess whether the evidence actually supports each finding. Missing evidence must remain visible. Suggested new factual claims require source support.

Core records

Organisation, Membership, Brand, SourceVersion, Rule, ApprovedFact, Asset, ContentRevision, ReviewRun, Finding, Comment, Decision and AuditEvent. Each tenant-owned record carries an organisation identifier. Findings reference a content revision and source version; decisions reference the same revision.

Use immutable submitted revisions, permission checks on every object lookup, atomic approval transitions and an expected-revision check to prevent approving stale content. When a source changes, mark affected approvals stale and require re-review before a new export.

05 | Responsible design and human control

Confidentiality and security

- **Minimise data:** Use synthetic demonstration content. Collect only the documents and context necessary for a review; redact personal information from research notes.
- **Restrict access:** Creators access their assigned work, reviewers access assigned reviews, and managers control source activation. Enforce permissions server-side and test cross-organisation isolation.
- **Protect uploads:** Validate file types and sizes, limit parsing time and decompression, disable active content, isolate extraction, and keep originals outside publicly served paths.
- **Protect stored and transmitted material:** Plan TLS for deployment, private storage, secret management and encrypted deployment volumes/backups. Verify these controls before any confidential pilot.
- **Control retention:** Show a configurable retention period and deletion workflow. Remove originals, extracted text, retrieval indexes and cached analysis together; document backup expiry and any minimal retained audit metadata.

AI transparency and reliability

NIST's risk profile informs the decision to evaluate model failure modes and preserve review evidence; it does not certify this product. [S3] Label model-generated findings, record model/prompt versions and processing mode, and disclose which content would leave the deployment boundary.

Before enabling an external provider, verify its current retention, training-use and regional processing terms. Do not promise confidentiality based only on the application's settings. Default to local processing, with semantic checks visibly unavailable if no approved model is configured.

Treat submitted content and retrieved passages as untrusted data, including instructions embedded in documents. The model receives no publishing, network-fetching or permission-changing tools. Test prompt-injection examples and schema failures. Timeouts or extraction failures must never become a clean review.

Meaningful human oversight

The system proposes corrections; creators choose changes. Only an assigned authorised reviewer can approve. Unsupported claims remain blocked until evidence is supplied, the claim is removed, or an appropriately authorised reviewer records a reasoned exception. An exception is not proof that a claim is true.

Separate source administration from content approval. Prevent self-approval in the pilot. Record the actor, revision, source snapshot, timestamp and decision rationale. Managers can review disputed findings and revise unclear rules without silently altering past evidence.

Accessible and proportionate feedback

Design the application for keyboard use, visible focus, descriptive labels, screen-reader announcements and severity indicators that include text. Test these behaviours directly. Evaluate tone feedback across different writing styles, avoid treating dialect as an error, and keep subjective findings advisory unless an explicit organisational rule applies.

These are design commitments and planned checks, not claims of completed security or accessibility certification.

06 | Delivery plan and expected impact

Development plan across all three rounds

CODE CATALYST will progress from problem validation to a working MVP and then a tested final demonstration. The round mapping below is the team's proposed delivery plan, based on the three stages in the supplied brief. Official deadlines and build-window dates remain subject to the organiser's announcements; product development starts only within the permitted window.

Round	Work package	Deliverables and exit evidence
Round 1: Proposal and validation	Define users, investigate evidence, specify workflow, architecture, responsibilities and evaluation. Conduct interviews where feasible.	One proposal PDF addressing all 11 requirements; research protocol and defined MVP scope. No invented interview results.
Round 2: Functional MVP	Build source management, intake, deterministic checks, AI adapter, bounded image checks, comments, revisions and approval.	Judge-accessible repository and installation/deployment guide; working draft-to-approval demo; architecture and dependency disclosures.
Round 3: Reliability and final submission	Evaluate held-out cases, gather feedback, fix failures, improve accessibility, test permissions and complete insights.	Final product, test results, limitations, contribution records and concise presentation; demo video or live demonstration as requested.

Implementation order and coordination

During Round 2, first establish the data model, permissions and versioned sources; then add findings and evidence; finally connect revision review and export. Round 3 begins from that working path and prioritises reliability before optional features.

Sanath coordinates integration and backend/AI work, Darshilkumar owns the interface and workflow, and Surya Prakash owns data preparation, evaluation and deployment. Full participant details and responsibilities appear in section 07. All three maintain contribution records and take part in testing.

Evaluation plan and targets

Create 60 labelled synthetic items: 20 for development and 40 held out for evaluation. Include clean content, all six detection categories, outdated/conflicting sources and extraction failures. Split by base content/template to reduce leakage. Have two reviewers independently label the held-out findings and adjudicate disagreements.

- **Finding quality:** Target at least 85% precision and 80% recall across labelled issues; report per category, sample counts and failures. Precision = correct flags / all flags; recall = detected labelled issues / all labelled issues.
- **Evidence:** Target 100% resolvable source references. Separately measure whether cited passages support the finding; target at least 90% on human review.
- **Workflow integrity:** All specified access-denial, stale-approval, self-approval and failure-state tests must pass.
- **User value:** Counterbalance manual-checklist and assisted review on matched drafts with the six participants. Target a 25% reduction in median review time without increased missed critical issues; report individual results and the small-sample limitation.

Expected impact: Creators should face fewer repeated corrections; reviewers should receive clearer evidence; managers should identify unclear or outdated guidelines. The targets above will test these benefits. No measured savings, ROI or accuracy results are claimed.

07 | Team and future scope

CODE CATALYST | Three-member team

Institution: RV College of Engineering, Bengaluru, inferred from the supplied institutional email addresses. Institution profile.

SANATH REDDY | Team leader

Email: sanathreddy.is24@rvce.edu.in

Responsibilities: Lead planning, architecture and integration; implement backend APIs, source versioning, deterministic checks and the AI/retrieval adapter; coordinate reviews and final submission. Own the technical explanation and evidence-grounding demonstration.

PATEL DARSHILKUMAR JYOTINDRAKUMAR | Team member

Email: pdjkumar.is24@rvce.edu.in

Responsibilities: Build the React interface, content intake, finding panels, comments, revision comparison, approval screens and insights dashboard. Own keyboard accessibility and browser workflow tests; prepare the product walkthrough with the team.

SATHI SURYA PRAKASH REDDY | Team member

Email: sathisuryapr.is24@rvce.edu.in

Responsibilities: Prepare synthetic brand packs and labelled evaluation cases; implement document/image extraction and bounded visual checks; test permissions, failure states and reliability. Own deployment packaging, evaluation results and the limitations log.

Shared responsibilities: Conduct user research, review each other's work, test the integrated product, maintain source/tool disclosures and record actual contributions. These are planned responsibilities, not claims of already completed engineering work. Sanath is the primary team contact.

Future scope after the event

Horizon	Planned extension and decision gate
First 1-2 months	Pilot with two willing teams; refine noisy findings, source ownership and retention controls. Proceed only with consent and acceptable evaluation results.
Months 3-4	Add Google Docs, Figma or CMS integrations based on pilot demand; evaluate multilingual guidance and stronger logo/layout checks using labelled examples.
Months 5-6 and beyond	Explore video/audio review, multi-brand workspaces, enterprise SSO and configurable private deployment. Validate security, accuracy and operating costs before expansion.

These are post-event planning horizons, not delivery commitments. Human approval, source-linked evidence and revision history remain core controls as formats and integrations expand.

08 | Submission coverage and references

One PDF | All 11 requested topics

Requirement	Location
1. Understanding of the problem statement	Opening summary
2. Users affected by the problem	Section 01
3. Proposed solution	Opening summary; section 02
4. Complete product workflow	Section 03
5. Main features intended for development	Section 02
6. Planned technology stack	Section 04
7. Responsibilities of each team member	Section 07
8. Development plan across all three rounds	Section 06
9. Expected impact	Section 06
10. Future scope after the event	Section 07
11. Team and participant details	Section 07

Submission details

Team name, all three participant names and emails, team leadership and planned responsibilities are included. Institution is inferred from the supplied email domain. Student/registration IDs and phone numbers were not supplied and are not invented; provide them separately if the organiser's submission form requires them.

The round plan is proposed; official dates and any extra organiser requirements must be checked before submission. Interviews, product development and measured outcomes remain planned.

Tools, reuse and assistance disclosure

This proposal was prepared with OpenAI Codex assistance for research synthesis, drafting and document preparation. Public sources informed the rationale; there was no product code, model training, user-data collection or external analysis of confidential brand materials. Python ReportLab was used to format the PDF.

The LumaDesk scenario and proposed evaluation data are synthetic. The existing pendulum research project in the surrounding workspace is unrelated and is not claimed as work on this entry. Record future reused code, templates, dependency licences, exact model/API identifiers and each member's actual work during the permitted build window.

References

[S1] Content Marketing Institute / MarketingProfs. “B2B Content Marketing Benchmarks, Budgets, and Trends: Outlook for 2025.” Published 9 October 2024. Survey limitations and sample described on the source page. Read the research.

[S2] Mailchimp. “Voice and Tone,” Content Style Guide. Used as a public example of contextual brand guidance, not copied as this product's guideline pack. Read the guidance.

[S3] NIST. “Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile,” NIST AI 600-1, July 2024. Publication and report.